TECHNICAL ARCHITECTURE & SERVICES

TeleSphere BSS/OSS Suite — Platform Reference Architecture Document

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This document is the authoritative technical reference for NexWave's TeleSphere BSS/OSS Suite. It serves as the primary knowledge base artifact for RFP automation — enabling automated matching of platform capabilities, standards compliance, SLA metrics, integration patterns, and technical specifications against telecom operator RFP requirements.

1. Platform Architecture Overview

1.1 Design Principles

TeleSphere BSS Suite v7.2 is built on the following foundational architectural principles, all aligned to Tier-1 telecom operator requirements:

- Cloud-Native: All services containerized (Docker/Helm); deployed on Kubernetes (EKS); infrastructure-as-code via Terraform
- Event-Driven Microservices: Domain-Driven Design (DDD) architecture; Apache Kafka v3.6 as central event bus (800K+ events/second capacity)
- API-First: TM Forum Open API-compliant REST and AsyncAPI; all services expose TM Forum-standard APIs natively
- Active-Active Multi-Region: Mumbai primary + Hyderabad DR; automated geo-failover; no single point of failure
- Horizontal Scalability: Horizontal Pod Autoscaler (HPA) for Kubernetes; tested to 300M subscribers without re-architecture
- Observability: Prometheus + Grafana (metrics); ELK stack (log aggregation); Jaeger (distributed tracing); Splunk SIEM (security)

1.2 Platform Component Map

BSS Module	Version	Key Standard	Production Scale
TeleSphere OCS	v7.2	3GPP Rel-17 Ro/Gy/Rf + Nchf	280M subs; 1.5M TPS (Stellar Mobile)
TeleSphere Billing Engine	v7.2	TMF676 + GST e-invoicing	280M accounts; 10M bill run in 4.2 hrs
TeleSphere CRM	v5.1	TMF632; SID-aligned data model	120M subscribers (AsiaNet Wireless)
TeleSphere OMS	v4.3	TMF622; BPMN (Camunda)	94.7% no-touch automation (AsiaNet)
TeleSphere NIM	v3.8	TMF637 + NETCONF/YANG	470,000+ nodes under management
TeleSphere RA/FM	v6.2	Apache Flink; LSTM ML models	<30 min leakage detection; <3 min fraud block
TeleSphere Self-Care	v4.0	WCAG 2.1 AA; PWA + iOS/Android	97.2% NLP chatbot intent accuracy
TeleSphere SliceOrchestrator	v2.1	3GPP TS 28.530; GSMA NG.116	Live at Crescent Telecom (5G SA — Q1 2024)

2. Online Charging System (OCS) — Technical Specification

2.1 Standards Compliance

Requirement	Status	Detail
3GPP Release 17 Ro/Gy/Rf interfaces	**COMPLIANT**	Ro (online charging), Gy (policy & charging), Rf (offline CDR); all 3GPP Rel-17 interfaces natively supported
5G SA Nchf interface (network slice charging)	**COMPLIANT**	5G SA Nchf interface live at Crescent Telecom (Q1 2024); network slice usage-based charging in production
OCS Rating Latency P99 <50ms at 1.5M TPS	**COMPLIANT**	<38ms P99 demonstrated at Stellar Mobile production environment (1.5M TPS sustained load)
Peak Throughput 1.5M TPS guaranteed	**COMPLIANT**	2M TPS load-tested; 1.5M TPS production SLA contractually guaranteed
Subscriber Scale to 300M	**COMPLIANT**	280M live at Stellar Mobile; tested to 300M; horizontal scaling without re-architecture
Prepaid real-time balance deduction	**COMPLIANT**	Real-time deduction, reservation, and reauthorization; <38ms P99
Postpaid convergent CDR-based billing	**COMPLIANT**	CDR-based mediation; convergent bill generation; postpaid+prepaid unified engine
Enterprise contract billing + CPQ	**COMPLIANT**	Contract billing, CPQ integration, volume discounts, SLA-based credits for enterprise accounts
OCS Availability: 99.999% (Five-Nines)	**COMPLIANT**	99.9997% achieved at Stellar Mobile over 12 months post-go-live; contractually committed

2.2 Data Layer Architecture

- Primary store: Apache Cassandra (distributed NoSQL) — high-throughput OCS balance management; sub-millisecond read/write P99
- Session state: Redis Cluster — real-time session reservation and balance lock management
- CDR mediation: Apache Kafka → custom mediation engine → PostgreSQL for billing input
- Anti-pattern: OCS and Billing use separate data stores — no OCS billing back-pressure during peak charging periods

3. BSS Module Specifications

3.1 CRM — Customer Management

Capability	TeleSphere CRM v5.1 Specification
Subscriber Segments	Consumer prepaid, postpaid, SME, enterprise — unified single CRM
360-Degree View	Omnichannel interaction history: call center, web, app, retail POS — unified timeline
Screen Load Time (P99)	<1.4 seconds P99 in production (RFP target: <2 seconds) — compliant
AI Agent Assist	Real-time sentiment analysis; suggested responses; automated case routing — TeleSphere AI engine
Churn Prediction	ML-based churn prediction; 30-day propensity score; next-best-action recommendations

Capability	TeleSphere CRM v5.1 Specification
Legacy Migration	Salesforce-compatible data model; Siebel v8.x bidirectional sync adapter; history migration tooling
TM Forum Compliance	TM Forum SID-aligned data model; TMF632 Party Management API natively exposed

3.2 Order Management System (OMS)

Capability	TeleSphere OMS v4.3 Specification
Architecture	Microservices OMS; BPMN-based workflow engine (Camunda v7); event-driven order orchestration
Order Types	45+ pre-built workflows: SIM swap, plan change, number porting, enterprise provisioning, 5G slice ordering
No-Touch Automation	94.7% no-touch order completion (AsiaNet Wireless Q1 2025); RFP target >90%
SIM Swap / Plan Change SLA	<2 hours average fulfillment (RFP target: <4 hours) — compliant with 50% headroom
New Connection SLA	<4 hours including SIM provisioning and network activation
Enterprise Order SLA	Complex CPQ + contract billing orders: <8 hours
API Compliance	TMF622 Product Order Management natively exposed

3.3 Revenue Assurance & Fraud Management

Capability	TeleSphere RA/FM v6.2 Specification
RA Engine	Real-time stream processing (Apache Flink); sub-30-minute leakage detection
Data Sources Reconciled	OCS events, mediation records, billing invoices, network CDRs — all 4 reconciled in real-time
Fraud Detection	Hybrid rule-based + ML model ensemble; LSTM-based anomaly detection
SIM Clone Alert-to-Block	<3 minutes in production at AsiaNet (RFP target: <5 minutes) — compliant with 40% headroom
ML Model Retraining	Daily automated retraining; false positive rate <0.5% (production-validated)
SUBEX RAG Migration	Pre-built migration toolkit; RA history preserved; no manual data re-entry
TRAI Regulatory Reports	Automated QoS and revenue report generation; no manual intervention required

4. OSS Module Specifications

4.1 Network Inventory Management (NIM)

- Vendor support: Ericsson (ECS, Radio, Core), Nokia (ADS, AirScale), Huawei (SDP, Enterprise), ZTE — pre-built adapters for all 4 major vendors
- Discovery protocols: SNMP v2c/v3, NETCONF/YANG, TR-069, SSH, REST APIs — all natively supported
- Topology views: Physical, logical, and service topology layers with live operational status

- Shadow inventory elimination: Automated reconciliation; decommissions Excel-based shadow systems within 90 days of go-live
- TMF637 Product Inventory Management API — natively exposed; real-time inventory queries

4.2 Service Provisioning

Capability	Specification
Provisioning Protocols	NETCONF/YANG, REST, SOAP, CLI templating — all four supported natively
SIM Provisioning SLA P95	<45 seconds P95 in production (RFP target: <60 seconds) — compliant
Rollback Capability	Automated rollback on provisioning failure; <30-second rollback execution time
5G NSA/SA Provisioning	AMF, SMF, UPF network function configuration via 5GC Service-Based Interface (SBI) APIs
eSIM / eUICC	RSP-compliant eSIM profile management; SGP.22 specification; remote provisioning capable

4.3 5G Network Slice Management

TeleSphere SliceOrchestrator v2.1 is production-deployed at Crescent Telecom (Saudi Arabia) since Q1 2024, making NexWave one of the first vendors globally with live 5G SA slice management in production.

- E2E slice lifecycle: Design → Order → Instantiate → Assure → Modify → Terminate — full lifecycle managed by TeleSphere
- Standards: GSMA NG.116 compliant; 3GPP TS 28.530 network slice management standard
- Slice types: eMBB (consumer broadband), URLLC (enterprise low-latency), mMTC (IoT/LPWA) — all 3 types in production
- Enterprise self-service: Slice ordering API exposed via TeleSphere Enterprise Portal; enterprise customers can order slices via portal
- Slice assurance: Real-time KPI monitoring per slice; SLA violation alerts within <2 minutes to enterprise customer portal
- Billing integration: 5G SA Nchf interface for usage-based slice billing; per-slice CDRs; convergent invoice generation

5. TM Forum Open API Compliance Matrix

TM Forum API	Status	TeleSphere Module	Notes
TMF620 — Product Catalog Mgmt	**COMPLIANT**	TeleSphere Catalog API v4.0	
TMF622 — Product Order Mgmt	**COMPLIANT**	TeleSphere OMS exposes TMF622 natively	
TMF632 — Party Management	**COMPLIANT**	CRM party model TM Forum SID-aligned	
TMF637 — Product Inventory	**COMPLIANT**	Real-time product inventory API	
TMF641 — Service Order Mgmt	**COMPLIANT**	OSS provisioning via TMF641	
TMF666 — Account Management	**COMPLIANT**	Enterprise account & billing APIs	

TM Forum API	Status	TeleSphere Module	Notes
TMF642 — Alarm Management	COMPLIANT	Fault Management module	
TMF724 — AI/ML Management	PARTIAL	AI/ML API exposure roadmap Q3 2026; current AI/ML used internally only	
TM Forum Frameworx Level	COMPLIANT	Level 4 certified — highest level; certificates in Appendix B	

6. Integration Architecture

6.1 Event Streaming Infrastructure

- Apache Kafka v3.6 — central event bus; minimum 800,000 events/second capacity (exceeds RFP requirement of 500K events/second)
- Confluent Schema Registry — Avro schema governance for all event types; schema evolution managed without breaking changes
- Pre-built Kafka connectors for all 40+ Vi integration points

6.2 Key Integration Catalog

Integration Target	Approach	Status / Reference
NPCI / UPI Payment Gateway	Pre-built adapter; certified for UPI 2.0, IMPS, NACH	Live at 3 Indian clients
Razorpay & Paytm	Payment aggregator SDKs — integrated	Live at 3 Indian clients
GSTN / IRP e-Invoicing	Certified IRN generation; CBIC-compliant	Live at BrightPath Mobile
Ericsson BSCS / ECS	Pre-built migration toolkit; subscriber data extraction certified	Used at AsiaNet (120M migration)
Siebel CRM v8.x	Bidirectional sync adapter; history migration tooling	Ready; tested against Siebel v8.1
Salesforce	Native Salesforce connector; real-time event sync	Live at 2 operator clients
SUBEX RAG	Data migration bridge; historical RA data preserved	Pre-built; ready for Vi SUBEX RAG v6.2
TRAI Reporting Automation	Automated QoS and regulatory report generation	No downtime during migration; no manual fallback
All 11 Indian CDR Exchange	Pre-built adapters for all 11 Indian operator interconnects	Ready for Vi's 22 telecom circles

7. Cloud Infrastructure Specification

7.1 Deployment Architecture

Infrastructure Component	Specification
Primary Region	AWS ap-south-1 (Mumbai) — MeitY-empanelled; ISO 27001; SOC 2 Type II

Infrastructure Component	Specification
DR Region	AWS ap-south-2 (Hyderabad) — active-active configuration; not cold standby
Container Orchestration	AWS EKS (Kubernetes); all 32 TeleSphere service components as containerized pods
Infrastructure as Code	Terraform for all provisioning; Helm charts for all service components; GitOps via ArgoCD
Auto-Scaling	Kubernetes HPA tuned for BSS traffic patterns (billing batch peaks vs. real-time OCS peaks)
Geo-Failover	Automated Route 53 health checks; RPO <15 min; RTO <1 hour; tested every 6 months
CDN (Self-Care Portal)	AWS CloudFront — global CDN for self-care portal; 99.95% availability SLA
Data Residency	100% in India; ap-south-1 + ap-south-2 only; no cross-border egress; contractually guaranteed

8. SLA Technical Commitments

SLA Metric	RFP Requirement	NexWave Commitment	Production Evidence
OCS/Charging Availability	99.999%	99.999% (contractual)	99.9997% at Stellar Mobile (12 months)
CRM/OMS Availability	99.99%	99.99% (contractual)	Active-active; 2 live references
Self-Care Portal Availability	99.95%	99.95% (contractual)	CloudFront CDN + multi-AZ
OCS Rating Latency P99	<50ms at 1.5M TPS	<50ms (contractual)	<38ms at Stellar Mobile (P99 production)
SIM Provisioning P95	<60 seconds	<60s (contractual)	<45 seconds P95 in production
Bill Run (10M accounts)	<6 hours	<6 hours (contractual)	4.2 hours tested (10M accounts)
Fraud Alert-to-Block	<5 minutes	<5 minutes (contractual)	<3 minutes at AsiaNet Wireless
Critical Incident Response	<15 minutes	<15 minutes (contractual)	24x7 NOC with automated paging
RPO (Disaster Recovery)	<15 minutes	<15 minutes (contractual)	Active-active; async replication <5s lag
RTO (Disaster Recovery)	<1 hour	<1 hour (contractual)	Automated Route 53 failover; tested 6-monthly

9. Data Migration Technical Framework

TeleSphere Migration Factory — Technical Methodology

NexWave's TeleSphere Migration Factory is a validated, tool-driven framework used in 6 major BSS transformations. Extract → Cleanse → Transform → Load → Validate → Reconcile → Cutover.

Migration Aspect	Specification	Evidence
Subscriber Volume	215M records (Siebel CRM + Ericsson ECS)	AsiaNet 120M reference (certified zero disruption)
Extraction Tool	TeleSphere DataBridge — certified connectors for Siebel v8.x and Ericsson ECS	Certified; used in 4 prior Ericsson/Siebel migrations
Data Cleansing	Automated deduplication, PII validation, address normalization, MSISDN reconciliation	Built-in DataBridge workflow
Transformation	SID-compliant data model mapping; 100% field-level audit trail	TMF632 SID compliance
Validation	Triple-check reconciliation: source count vs. landing zone vs. target	Standard Migration Factory protocol
CDR History	24 months migrated and queryable; AWS S3 Glacier archive for older CDRs	TRAI CDR retention compliance
Parallel Run	60-day mandatory parallel run per RFP; legacy and new systems live simultaneously	Used at AsiaNet, Stellar Mobile
Rollback Capability	Instant rollback to legacy during 60-day parallel run; no subscriber impact	Contractually committed

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